

SQL Analysis Report

The screenshot shows the SQL Server Enterprise Manager interface. The left pane displays the 'SCHEMAS' tree with 'employee_attrition_db' expanded, showing the 'employee_attrition' table. The right pane shows 'Query 1' with the following SQL code:

```
1 • SELECT * FROM employee_attrition_db.employee_attrition;
2 -- Question 1 What is the overall employee attrition rate?
3 • SELECT
4 ROUND(
5 SUM(CASE WHEN Attrition='Yes' THEN 1 ELSE 0 END)*100.0
6 / COUNT(*),2
7 ) AS Attrition_Rate
8 FROM employee_attrition;
```

The 'Information' pane at the bottom left shows the table 'employee_attrition' with columns: Age, Attrition, BusinessTravel, DailyRate, etc. The 'Result Grid' at the bottom right shows a single row with the value '16.12' for the 'Attrition_Rate' column.

Attrition_Rate
16.12

- 1) The overall employee attrition rate is **16.12%**, indicating that approximately 16 out of every 100 employees leave the organization. This suggests a moderate turnover rate and highlights the need for proactive employee retention strategies.

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```
1 • SELECT * FROM employee_attrition_db.employee_attrition;
2 -- Question 2 Attrition by Department
3 • SELECT
4 Department,
5 COUNT(*) AS Total_Employees,
6 SUM(CASE WHEN Attrition='Yes' THEN 1 ELSE 0 END) AS Attrition_Count
7 FROM employee_attrition
8 GROUP BY Department
9 ORDER BY Attrition_Count DESC;
```

The 'Information' pane at the bottom left shows the table 'employee_attrition' with columns: Age, Attrition, BusinessTravel, DailyRate, Department, DistanceFromHome, Education, EducationField, etc. The 'Result Grid' at the bottom right shows the following data:

Department	Total_Employees	Attrition_Count
Research & Development	961	133
Sales	446	92
Human Resources	63	12

- 2) The **Research & Development department** recorded the highest attrition count (**133 employees**), followed by Sales (**92 employees**) and Human Resources (**12 employees**). Retention initiatives should primarily focus on the Research & Development function.

The screenshot displays the SQL Server Enterprise Manager interface. On the left, the 'SCHEMAS' pane shows the 'employee_attrition_db' database with the 'employee_attrition' table selected. The 'Information' pane at the bottom left lists the columns of the 'employee_attrition' table: Age, Attrition, BusinessTravel, DailyRate, Department, DistanceFromHome, Education, EducationField, EmployeeCount, EmployeeNumber, EnvironmentSatisfaction, Gender, HourlyRate, JobInvolvement, and JobLevel. The main query window shows a SQL query for 'Query 1' in 'employee_attrition.csv'. The query is as follows:

```
1 • SELECT * FROM employee_attrition_db.employee_attrition;
2 -- Question 3 Attrition by Job Role
3 • SELECT
4 JobRole,
5 COUNT(*) AS Total_Employees,
6 SUM(CASE WHEN Attrition='Yes' THEN 1 ELSE 0 END) AS Attrition_Count
7 FROM employee_attrition
8 GROUP BY JobRole
9 ORDER BY Attrition_Count DESC;
```

The 'Result Grid' at the bottom right shows the results of the query, sorted by 'Attrition_Count' in descending order. The results are as follows:

JobRole	Total_Employees	Attrition_Count
Laboratory Technician	259	62
Sales Executive	326	57
Research Scientist	292	47
Sales Representative	83	33
Human Resources	52	12
Manufacturing Director	145	10
Healthcare Representative	131	9
Manager	102	5
Research Director	80	2

- 3) **Laboratory Technicians (62), Sales Executives (57), and Research Scientists (47)** experienced the highest employee turnover. These roles should be prioritized for engagement and retention programs.

Navigator

SCHEMAS

Filter objects

demodb

employee_attrition_db

Tables

employee_attrition

Columns

Indexes

Foreign Keys

Triggers

Views

Stored Procedures

Functions

sakila

sys

Tables

Views

Administration

Schemas

Information

Table: **employee_attrition**

Columns:

Age

Attrition

BusinessTravel

DailyRate

Department

DistanceFromHome

Education

Query 1 employee_attrition.csv employee_attrition

Limit to 1000 rows

```

1 • SELECT * FROM employee_attrition_db.employee_attrition;
2 -- Question 4 Attrition by Gender
3 • SELECT
4 Gender,
5 COUNT(*) AS Total_Employees,
6 SUM(CASE WHEN Attrition='Yes' THEN 1 ELSE 0 END) AS Attrition_Count
7 FROM employee_attrition
8 GROUP BY Gender;

```

Result Grid

Filter Rows:

Export: Wrap Cell Content: [FA](#)

Gender	Total_Employees	Attrition_Count
Female	588	87
Male	882	150

- 4) Male employees accounted for **150 attrition cases**, while female employees accounted for **87 attrition cases**. This indicates a higher turnover volume among male employees and may require further analysis of role distribution and workplace factors.

Navigator

SCHEMAS

Filter objects

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Views

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Functions

sakila

sys

Tables

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Administration

Schemas

Information

Table: **employee_attrition**

Columns:

Age

Attrition

BusinessTravel

DailyRate

Department

DistanceFromHome

Query 1 employee_attrition.csv employee_attrition

Limit to 1000 rows

```

1 • SELECT * FROM employee_attrition_db.employee_attrition;
2 -- Question 5 Overtime Impact
3 • SELECT
4 OverTime,
5 COUNT(*) AS Employees,
6 SUM(CASE WHEN Attrition='Yes' THEN 1 ELSE 0 END) AS Attrition_Count
7 FROM employee_attrition
8 GROUP BY OverTime;

```

Result Grid

Filter Rows:

Export: Wrap Cell Content: [FA](#)

OverTime	Employees	Attrition_Count
Yes	416	127
No	1054	110

- 5) Employees working overtime showed significantly higher attrition (**127 employees**) compared to employees who did not work overtime (**110 employees**), despite the overtime group being much smaller. This suggests that excessive workload is a major contributor to employee turnover.

The screenshot displays a database management interface. On the left, the 'SCHEMAS' pane shows a tree view with 'employee_attrition_db' expanded, revealing tables, columns, indexes, foreign keys, triggers, views, stored procedures, and functions. The 'employee_attrition' table is selected. Below this, the 'Table: employee_attrition' section lists columns: Age, Attrition, BusinessTravel, DailyRate, Department, DistanceFromHome, Education, EducationField, and EmployeeCount.

The main pane shows 'Query 1' with the following SQL code:

```
1 • SELECT * FROM employee_attrition_db.employee_attrition;
2 -- Question 6 Work-Life Balance Impact
3 • SELECT
4   WorkLifeBalance,
5   COUNT(*) AS Employees,
6   SUM(CASE WHEN Attrition='Yes' THEN 1 ELSE 0 END) AS Attrition_Count
7 FROM employee_attrition
8 GROUP BY WorkLifeBalance
9 ORDER BY WorkLifeBalance;
```

Below the query, the 'Result Grid' shows the following data:

	WorkLifeBalance	Employees	Attrition_Count
1	80	25	
2	344	58	
3	893	127	
4	153	27	

- 6) Employees with lower work-life balance ratings exhibited higher attrition levels. Employees rated at level **3** recorded the highest attrition count (**127 employees**), indicating that work-life balance is an important factor influencing employee retention.

Navigator: SCHEMAS

Filter objects

demodb

employee_attrition_db

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employee_attrition

Columns

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Foreign Keys

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Views

Stored Procedures

Functions

sakila

sys

Tables

Views

Administration Schemas

Information

Table: employee_attrition

Columns:

Age

Attrition

BusinessTravel

DailyRate

Department

DistanceFromHome

Education

EducationField

EmployeeCount

Query 1 employee_attrition.csv employee_attrition

Limit to 1000 rows

```

1 • SELECT * FROM employee_attrition_db.employee_attrition;
2 -- Question 7 Job Satisfaction Analysis
3 • SELECT
4 JobSatisfaction,
5 COUNT(*) AS Employees,
6 SUM(CASE WHEN Attrition='Yes' THEN 1 ELSE 0 END) AS Attrition_Count
7 FROM employee_attrition
8 GROUP BY JobSatisfaction
9 ORDER BY JobSatisfaction;

```

Result Grid

Filter Rows:

Export: Wrap Cell Content:

	JobSatisfaction	Employees	Attrition_Count
1	1	289	66
2	2	280	46
3	3	442	73
4	4	459	52

- 7) Employees with lower job satisfaction levels experienced higher attrition. Job satisfaction level 3 recorded the highest attrition count (**73 employees**), highlighting the importance of employee engagement and satisfaction initiatives.

Navigator: SCHEMAS

Filter objects

demodb

employee_attrition_db

Tables

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Functions

sakila

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Tables

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Administration Schemas

Information

Table: employee_attrition

Columns:

Age

Attrition

BusinessTravel

DailyRate

Department

DistanceFromHome

Education

EducationField

EmployeeCount

Query 1 employee_attrition.csv employee_attrition

Limit to 1000 rows

```

1 • SELECT * FROM employee_attrition_db.employee_attrition;
2 -- Question 8 Attrition by Marital Status
3 • SELECT
4 MaritalStatus,
5 COUNT(*) AS Employees,
6 SUM(CASE WHEN Attrition='Yes' THEN 1 ELSE 0 END) AS Attrition_Count
7 FROM employee_attrition
8 GROUP BY MaritalStatus;

```

Result Grid

Filter Rows:

Export: Wrap Cell Content:

	MaritalStatus	Employees	Attrition_Count
1	Single	470	120
2	Married	673	84
3	Divorced	327	33

- 8) Employees with lower job satisfaction levels experienced higher attrition. Job satisfaction level **3** recorded the highest attrition count (**73 employees**), highlighting the importance of employee engagement and satisfaction initiatives.

The screenshot displays a database management interface. On the left, the 'SCHEMAS' pane shows a tree view with 'employee_attrition_db' expanded, revealing the 'employee_attrition' table. The 'Information' pane at the bottom left lists the columns of the 'employee_attrition' table: Age, Attrition, BusinessTravel, DailyRate, Department, DistanceFromHome, and Education. The main query editor shows a SQL query: `SELECT * FROM employee_attrition_db.employee_attrition; -- Question 9 Average Salary by Attrition SELECT Attrition, ROUND(AVG(MonthlyIncome),2) AS Avg_Salary FROM employee_attrition GROUP BY Attrition;`. The 'Result Grid' at the bottom right shows the output of the query, which is a table with two columns: 'Attrition' and 'Avg_Salary'. The data is as follows:

Attrition	Avg_Salary
Yes	4787.09
No	6832.74

- 9) Employees who left the organization had an average monthly income of approximately **₹4,787**, whereas retained employees earned approximately **₹6,833**. This indicates that compensation may play a significant role in employee retention decisions.

Navigator: SCHEMAS

Filter objects

- demodb
 - employee_attrition_db
 - Tables
 - employee_attrition
 - Columns
 - Indexes
 - Foreign Keys
 - Triggers
 - Views
 - Stored Procedures
 - Functions
 - sakila
 - sys
 - Tables
 - Views

Administration Schemas

Information

Table: **employee_attrition**

Columns:

- Age
- Attrition
- BusinessTravel
- DailyRate
- Department
- DistanceFromHome
- Education
- EducationField
- EmployeeCount
- EmployeeNumber
- EnvironmentSatisfaction

Query 1 employee_attrition.csv employee_attrition x

Limit to 1000 rows

```

1 • SELECT * FROM employee_attrition_db.employee_attrition;
2 -- Question 10 Top Risk Job Roles
3 • SELECT
4   JobRole,
5   SUM(CASE WHEN Attrition='Yes' THEN 1 ELSE 0 END) AS Attrition_Count
6 FROM employee_attrition
7 GROUP BY JobRole
8 ORDER BY Attrition_Count DESC
9 LIMIT 5;

```

Result Grid

JobRole	Attrition_Count
Laboratory Technician	62
Sales Executive	57
Research Scientist	47
Sales Representative	33
Human Resources	12

Export: | Wrap Cell Content: | Fetch rows:

10) The highest-risk job roles for attrition are **Laboratory Technician, Sales Executive, Research Scientist, Sales Representative, and Human Resources**. HR should prioritize these roles for retention planning and employee engagement initiatives